

Age-Differentiated Use of Social Platforms

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There are now numerous websites on the Internet that provide social-media services to users. People use these platforms differently. In this paper, we study the difference in social platform usage between older and younger adults. By analysing a survey we conducted of 101 older users (60+) and 32 younger users (between 19-25 years), we have been able to measure not only a difference in the frequency of usage across several social platforms, but also a difference in purposes. This is the first differential, intergenerational assessment across social platforms that we are aware of.

CCS Concepts: • **Information systems** → **Social networks**; • **Social and professional topics** → **Seniors**; • **Human-centered computing** → **Social networking sites**; *Social network analysis*; *User studies*.

Additional Key Words and Phrases: older adults, social-media platforms, survey

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1 Introduction

Today we rely on the internet for almost all of our needs: from shopping for essential items and finding health information (physical needs) to communicating with family and friends and entertainment (emotional needs). As a result, we use various online platforms every day. At the same time, the use of these platforms is not restricted to a specific age group; in some respects, everyone, young or old, needs to use them.

It is widely believed, however, that there is a technological gap between the old and the young [8, 10]. There have been previous studies that attempt to better understand this gap. Facebook has been particularly mesmerising in this respect [3, 5, 6]. But is there any differentiation in which of the several available social platforms older or younger users will choose for particular tasks? Have they even heard of all of them?

This paper presents the results of a survey we conducted of both young and old users in order to investigate the differences between them in terms of brand awareness, brand exposure (whether they have tried them), and frequency of use.¹ We also report statistical significance tests, coupled with an analysis and discussion. Our data and analysis should be useful to designers of social platforms, market researchers and social scientists.

¹We also collected data on task specificity, which are omitted here because of space limitations.

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Table 1. Awareness of Different Social Platforms

	Facebook	X/Twitter	WhatsApp	Telegram	Instagram	Pinterest	Reddit	LinkedIn	Youtube	TikTok	Amazon	Ebay	Twitch	Medium	Google	Wattpad	Yahoo	Bing	ChatGPT
Old	98%	79%	66%	45%	77%	64%	62%	71%	95%	65%	91%	74%	28%	25%	90%	11%	72%	57%	66%
Young	100%	97%	75%	50%	97%	75%	88%	69%	97%	91%	94%	84%	88%	9%	97%	59%	72%	63%	88%
p_{χ^2}	.42	.02	.36	.59	.01	.26	.01	.78	.66	.01	.63	.24	.00	.06	.23	.00	.96	.61	.02

2 Study Apparatus and Participants

We designed a web-based survey using the LimeSurvey[4] tool to collect these data. Subjects fill out a brief demographic questionnaire (occupation, education, and home/mobile internet access), followed by a few questions about digital gadgets that they use or have access to, and the tasks that they use them for.

We then turned to social platforms. Here, "social platform," refers to all services that enable users to share their opinion (as posts or comments), even if they are not dedicated social networks in the same manner as Facebook or X/Twitter. This part consists of four questions, covered in Sections 4.1–4.4.

We separately recruited two groups of participants through Prolific. The first group (101 participants, $\bar{x} = 65.5$, $\tilde{x} = 65$, $s = 4.65$) consisted of older adults who are over 60 years old, live in the UK, US, or Canada, and have a good command of English (the language of the survey). The second group (32, $\bar{x} = 23.1$, $\tilde{x} = 24$, $s = 1.86$) consisted of younger participants who were between 18 and 25 years old.

3 Results

3.1 Awareness of Social Platforms

Our first question aimed to see how many of the major social platforms are known to subjects, and whether there is any significant difference between older and younger people. Table 1 shows the answers to this question. Overall, young people are more familiar with social platforms (higher awareness in all cases). A χ^2 -squared test² reveals for which platforms the difference between young and old is significant. We can see that young users are more often aware of Wattpad, Twitch, TikTok, Reddit, Instagram, X/Twitter, and ChatGPT, although only the first two with statistical significance. These results are consistent with the small number of previously published reports that have enquired into awareness of single brands (versus studies of the effectiveness of social platforms in promoting awareness of non-social-platform brands, which are numerous). For example, a majority (85%) of Wattpad users are under 30 [1].

3.2 Exposure to Different Social Platforms

Our second question aimed to go one step further: how many people tried and then became a user of each social platform. The results of this question are presented in Table 2.

The data from this and the next question illustrate the distribution of users across different platform categories. We used two metrics, reported in Table 2, to compare these histograms between older and younger users. Earth Mover

²Green means significant even after Holm-Bonferroni correction. Yellow means that it would have been significant had we tested only that one platform. $\alpha = 0.05$.

Table 2. Exposure to Social Platforms

		Facebook	X/Twitter	WhatsApp	Telegram	Instagram	Pinterest	Reddit	LinkedIn	Youtube	Tiktok	Shopping	Search	Travel	Chatbot	News	Stream	MediaPub
Old	Use	79%	48%	36%	17%	44%	30%	47%	44%	95%	27%	89%	91%	51%	51%	70%	71%	16%
	No More	9%	12%	5%	7%	7%	13%	8%	14%	2%	8%	4%	6%	16%	4%	15%	11%	6%
	Tried	8%	13%	6%	11%	19%	22%	17%	15%	1%	18%	7%	1%	15%	16%	10%	6%	8%
	Heard	4%	26%	47%	46%	31%	33%	22%	24%	2%	47%	0%	2%	16%	23%	3%	11%	21%
	Not Heard	0%	2%	7%	20%	0%	3%	7%	4%	0%	1%	0%	0%	3%	7%	2%	1%	50%
Young	Use	66%	66%	28%	19%	72%	41%	72%	22%	94%	50%	88%	100%	31%	59%	56%	88%	19%
	No More	28%	19%	22%	13%	25%	13%	16%	25%	3%	9%	6%	0%	19%	16%	19%	9%	28%
	Tried	6%	6%	22%	9%	3%	13%	6%	16%	0%	22%	0%	0%	9%	9%	13%	3%	13%
	Heard	0%	9%	22%	41%	0%	31%	6%	34%	3%	19%	6%	0%	34%	16%	13%	0%	25%
	Not Heard	0%	0%	6%	19%	0%	3%	0%	3%	0%	0%	0%	0%	6%	0%	0%	0%	16%
EMD		0.19	0.61	0.42	0.15	0.75	0.23	0.81	0.42	0.01	0.77	0.02	0.12	0.57	0.44	0.32	0.43	0.58
q-value		0.81	0.00	0.00	0.93	0.00	0.62	0.00	0.00	1.00	0.00	1.00	0.94	0.00	0.00	0.23	0.00	0.00
PSI		0.95	0.64	0.66	0.04	6.66	0.09	1.70	0.26	0.17	0.57	2.39	1.56	0.29	1.48	0.52	2.23	0.77

Distance (EMD) is often used in different domains for comparing histograms[2].³ The Population Similarity Index (PSI)[9] determines how similar two populations are in respect of an attribute.⁴

Based on EMD, the more conservative of the two, we can see that for roughly two thirds of the platforms, there is a significant difference between young and old. The major exceptions, as shown by PSI, are Telegram and Pinterest. The Telegram case is interesting as 19% of younger participants reported that they had not heard about it, while another 41% had heard about it but had not tried it. These numbers are very similar to those for older participants. In addition, Youtube has very high usage among both groups (95% among old vs 94% among young).

3.3 Frequency of Using Social Platforms

The results of Table 2 show that the majority of users in both groups use most social platforms — in some cases, such as Youtube and Search, almost everyone. Our third question aimed to better compare younger and older subjects by looking at how frequently they use each of the platforms. Table 3 shows the distribution of the answers to this question.

Here, there is some disagreement between the two measures as to which platforms have demonstrated significant differences. They agree that 9 of them do; at least one thinks that 16 do. The interesting thing to note is that the explanation for these differences is not that younger people use the platforms more. In fact, the opposite is true for some platforms, e.g., Facebook (55% of the older group use Facebook each day, but 41% of the younger). This is consistent with previous reports that older people are more engaged with Facebook than younger people in respect of content sharing [6]. The same trend in frequency, however, holds for News (54% vs. 31%) and Telegram (14% vs. 6%).

3.4 Tasks Completed on Social Platforms

Our fourth question focuses on how both groups use different platforms for 12 classes of tasks that users might perform on a social platform: Friends and Family, Passing time, Self-expression, Business communication, Research, News, Reading books/periodicals, Watching movies/sports, Health and fitness, Travel planning, Online shopping, and Dating.

³We also calculated its associated q-value using the EMDomics[7] package, but we found their distribution to be suspiciously brittle — either very high or very low.

⁴Siddiqi[9] (page 137) instructs that the populations are similar if the PSI is less than 0.1, significantly different if the PSI is greater than 0.25, and marginally significant between 0.10 and 0.25. We have not yet attempted to convert these into *p*-values that can be corrected.

Table 3. Frequency of Using Different Social Platforms

		Facebook	X/Twitter	WhatsApp	Telegram	Instagram	Pinterest	Reddit	LinkedIn	Youtube	Tiktok	Shopping	Search	Travel	Chatbot	News	Stream	MediaPub
Old	Daily	55%	28%	28%	14%	23%	13%	24%	9%	70%	18%	19%	84%	4%	21%	54%	46%	3%
	Weekly	8%	9%	5%	2%	12%	8%	12%	15%	13%	7%	31%	4%	3%	16%	13%	18%	6%
	Monthly	11%	5%	1%	2%	8%	10%	12%	6%	8%	8%	28%	7%	13%	10%	10%	4%	8%
	Occasions	1%	1%	0%	4%	3%	6%	4%	6%	1%	4%	7%	0%	28%	1%	2%	3%	3%
	Irregular	6%	10%	5%	1%	10%	12%	9%	13%	4%	2%	10%	2%	20%	11%	6%	9%	7%
	NotUse	13%	22%	23%	23%	22%	25%	15%	34%	1%	23%	5%	1%	19%	19%	12%	5%	18%
	NoAccess	6%	26%	39%	55%	23%	27%	25%	18%	3%	39%	1%	2%	14%	23%	4%	16%	55%
Young	Daily	41%	41%	19%	6%	63%	16%	56%	9%	72%	44%	19%	91%	0%	31%	31%	53%	6%
	Weekly	9%	16%	0%	3%	9%	22%	19%	6%	16%	0%	25%	9%	9%	16%	22%	22%	3%
	Monthly	13%	16%	9%	9%	9%	6%	6%	9%	6%	6%	41%	0%	0%	13%	6%	16%	13%
	Occasions	3%	0%	0%	3%	3%	3%	3%	0%	0%	3%	6%	0%	28%	0%	6%	0%	0%
	Irregular	9%	3%	9%	3%	3%	13%	6%	19%	3%	6%	6%	0%	16%	25%	22%	9%	9%
	NotUse	19%	9%	38%	28%	6%	22%	6%	44%	0%	22%	0%	0%	31%	13%	9%	0%	34%
	NoAccess	6%	16%	25%	47%	6%	19%	3%	13%	3%	19%	3%	0%	16%	3%	3%	0%	34%
EMD	0.56	1.15	0.35	0.17	1.87	0.54	1.69	0.29	0.11	1.00	0.22	0.31	0.41	0.71	0.68	0.84	0.15	
q-value	0.00	0.00	0.00	0.93	0.00	0.00	0.00	0.46	0.94	0.00	0.62	0.30	0.00	0.00	0.00	0.00	0.93	
PSI	0.11	0.60	1.27	0.23	0.89	0.21	0.88	1.21	0.33	1.68	0.99	2.13	3.26	0.75	0.45	4.56	0.80	

For lack of space, the data are not shown here, but briefly, the majority of platforms were selected only for a few closely related tasks (such as passing time and news). On the one hand, for most tasks (connecting with friends and family, business communications, health, shopping, and dating), a χ^2 reveals no difference between older and younger. But there are significant differences in the selection of different platforms by different age groups. The prime example is TikTok, which is used significantly differently by the younger group in 4 tasks: Self-Expression, Research, News, and Watch. Reddit, which is used differently in 3 tasks: Passing Time, Self Expression, and News, is next.

4 Discussion

Our survey aimed to corroborate perceived differences between younger and older social media users in terms of their awareness of different online platforms and the extent to which they have tried them. Furthermore, it examined how frequently either group uses each platform and for which tasks they use them.

In part perhaps because of the sample bias in both age groups towards subjects who are comfortable with using computers, this divide seems not to be very large. For more than half of the platforms studied, there was no significant difference between younger and older subjects in terms of awareness. On the other hand, for a majority of platforms, a significantly low percentage of the older group either had not heard about a platform at all or had heard about it but not tried it. There are also groups of platforms, such as Shopping and News, that almost all of the older group had at least tried, but a significant portion of the younger group (6% for Shopping and 13% for News) had not tried.

This suggests that neither a lower frequency of use, nor even a lower percentage of exposure to different social platforms necessarily correlates with being older. They also do not correlate with having little awareness of them. It suggests that they are both the result of a deliberate decision against the platform, either because it is associated with unnecessary tasks, or because the need for the tasks that it does offer is already met by another platform.

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